

portfolio beyond the conventional holding period of 1–12 months. This helps us to understand the behaviour of the four sub-portfolios and illustrates which one contributes most to the time series reversal. The performance evaluation period is extended to a post-holding period of 13–60 months after the signal.¹⁴

The time horizon is divided into three parts: (i) the 12-month look-back period j prior to month 1 (see ranking period 1 in Figure 1); (ii) the holding period h ranging from month 1–12 (see ranking period 2 in Figure 1);¹⁵ and (iii) the 13–60 month post holding period. As before, the same two trend-following strategies—TSM and RSM—are performed. A rolling method is employed to generate and renew the trading signals every month.

Analogous to [Conrad and Yavuz \(2017\)](#), we decompose the trend-following portfolio into four subgroups: realised winner, realised loser, contrarian winner, and contrarian loser. First, based on each instrument’s signals formed in ranking period 1 ($j = 12$), we divide the entire portfolio into winners and losers. We then classify the winner and loser portfolios into the above-mentioned four subgroups by evaluating whether or not their profits are realised during ranking period 2 ($h = \{1, 2, 3, 6, 12\}$).¹⁶

More specifically, the realised winner represents a sub-portfolio with instruments that were past winners during period j and continue to generate positive returns during period h . The realised loser represents a sub-portfolio that consists of past losers during period j that continue to generate negative returns during period h . By contrast, the contrarian winner represents a sub-portfolio comprising past winners during period j , which then fail to generate positive returns during period h . The contrarian loser represents a sub-portfolio with past losers that go on to generate positive returns during period h . These four subgroups cover all the assets in the

¹⁴Post-holding period study of momentum strategies was also seen in [Jegadeesh and Titman \(2001\)](#), who tracked cross-sectional momentum performance up to 60 months after the formation of momentum signals.

¹⁵We leave $h = \{1, 2, 3, 6, 12\}$ as this is usual in calculating momentum profits, see, e.g., [Moskowitz et al. \(2012\)](#).

¹⁶Figure 1 illustrates how these four sub-portfolios are constructed.